ABHAY TIWARI

AI ENGINEER

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OBJECTIVE

Results-driven AI & Machine Learning Engineer with hands-on experience in data analysis, predictive modeling, and production deployment of machine learning and AI solutions. Proficient in Python, NLP, deep learning, and model optimization. Experienced in TensorFlow, PyTorch, Scikit-learn, and Pandas, with expertise in building scalable applications using Flask and REST APIs. Skilled in LangChain, LangGraph, LLMs, and RAG pipelines for intelligent agentic systems.

WORK EXPERIENCE

AI Research & Engineering Intern — UBS, Pune Aug 2026 – Present

- Built a hybrid personalized research recommendation engine combining Rule-Based scoring (freshness/recency/trending), Matrix Factorization, and Semantic Similarity, with XGBoost/CatBoost reranking for dynamic weight personalization.
- Designed a production-grade multi-table feature schema with affinity profiles, document metadata, and cold-start handling; resolved data quality issues with SMEs to surface high-signal features like RIC and ProductCode.
- Benchmarked agentic frameworks (LangChain, DeepAgents, Hermes) and built multi-agent pipelines for automated data validation, inconsistency detection, and stakeholder reporting workflows.

AI & Full Stack Developer Intern (*Remote*) — Infosys Springboard Oct 2025 – Dec 2025

- Built an AI-powered fraud detection system using Random Forest, achieving 91.3% accuracy across transaction data.
- Developed scalable FastAPI backend services with MongoDB and React.js for real-time analytics and decision support.
- Optimized Docker-based microservices with Redis caching, reducing API latency by 40% and improving scalability.

EDUCATION

Vellore Institute of Technology, Chennai Aug 2023 – Jul 2027
B.Tech in Computer Science (AI & ML) **CGPA: 9.36**

Sri Chaitanya Junior Kalasala, Hyderabad Jun 2021 – May 2023
Class XII — Board of Secondary Education Telangana **98.9%**

Shloka CBSE School Apr 2020 – May 2021
Class X — Central Board of Secondary Education **93.8%**

TECHNICAL SKILLS AND INTERESTS

Coding Languages	Python, JavaScript, Java, C, C++
AI & Data Science	Machine Learning, Deep Learning, NLP, TensorFlow, PyTorch, Scikit-learn
Agentic AI & LLM Systems	LangChain, LangGraph, LLMs, RAG Pipelines, Vector Databases, Multi-Agent Systems, Tool Calling, Workflow Orchestration
DevOps & MLOps	Docker, Kubernetes, CI/CD, Jenkins, Apache Kafka, Redis
Tools & Platforms	Git, GitHub, GitHub Actions, Postman, VS Code, Jupyter Notebook, Linux, AWS, Google Cloud, SQL, MongoDB
Core CS Concepts	Data Structures & Algorithms, OOPs, Operating Systems, DBMS, Computer Networks
Soft Skills	Problem Solving, System Design Thinking, Analytical Thinking, Team Collaboration, Communication, Adaptability

PROJECTS

Cortex Enterprise (FastAPI, React.js, MongoDB, Microservices)

- Built full-stack supply chain platform with real-time monitoring, analytics, and decision support.
- Developed REST APIs and integrated ML-based demand forecasting for inventory optimization.
- Designed scalable microservices architecture using FastAPI, React.js, and MongoDB.

Real-Time Logistics ML Pipeline (Apache Kafka, Machine Learning, Python, Docker)

- Built real-time ML pipeline using Apache Kafka for anomaly detection and predictive analytics.
- Developed dashboard for monitoring risk metrics and operational data streams.
- Deployed models via Flask APIs with Docker for scalable real-time processing.

Hybrid Semantic Search System (VectraCache) (FastAPI, FAISS, NLP, Docker)

- Built hybrid semantic search system using transformer embeddings and keyword-based ranking to improve search relevance.
- Implemented clustering-based query routing and semantic caching to reduce latency and optimize performance.
- Developed FastAPI service with FAISS indexing for efficient similarity search and scalable deployment using Docker.

CERTIFICATIONS AND ACHIEVEMENTS

Certifications: Generative AI with Watsonx – IBM (2025), Oracle Certified Generative AI Professional (2025), Oracle Cloud Database Services (2025)

Achievements: **Qualified GATE**, HackFest 2.0 (GDG IIT Delhi) Top 7, Sustainovate Hackathon 5th Position, Tutedude Buildathon 2025 Top 5, Frontend Battle 2.0 (IIT Bhubaneswar) Finalist